

Biological Transitions as Multi-Agent Realisations of the Generative Operator Pipeline in Topographical Orthogonal Generative Theory (TO/TOGT)

Version 2 — LaTeX source, complete references, and companion cross-reference added

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Abstract

We model collective biological transitions — including neural oscillations, HPA-axis stress response, circadian regulation, immune adaptation, and protein conformational change — as multi-agent realisations of the generative operator pipeline $G = U \circ F \circ K \circ C$ introduced in Topographical Orthogonal Generative Theory (TO/TOGT). Each agent undergoes local compression, curvature intensification, loss of injectivity, and stabilisation. The collective fixed points, contraction properties, and saturated pitchfork bifurcation behaviour follow directly from the fixed-point theory, contraction-mapping results, and algebraic separation theorem proved in the author’s earlier work [1] (HAL hal-05555216v1). The specific dm^3 numerical invariants ($T^* = 2\pi$, $\mu_{\max} = -2$, $\tau = 2$) are convenient modelling choices matching observed biological scales; they are not derived theorems of the operator algebra. No numerical simulations or unverified biological claims are made.

This article is part of the Principia Orthogona / Generative Contact Mechanics series and demonstrates a concrete multi-agent biological application of the TO/TOGT operator pipeline. A companion paper [3] specialises the framework to the *Drosophila* connectome, adding Python simulation code, Lean 4 proofs, and four figures.

Keywords: generative operator pipeline; TO/TOGT; multi-agent systems; biological transitions; dynamical systems; contraction mapping; saturated pitchfork bifurcation; connectomics; HPA axis; generative transitions

MSC codes: 37C25, 37D10, 37G10, 47H10, 92B20, 92C20

1 Relation to Prior Work

This article applies the mathematical results established in the author’s earlier preprint *Principia Orthogona, Volume One: The Mathematics of Generative Transitions* [1] (HAL hal-05555216v1). There the composite operator pipeline

$$G = U \circ F \circ K \circ C$$

was introduced together with complete proofs of fixed-point existence and uniqueness, global contraction, saturated pitchfork bifurcation, and the algebraic trace-bound/separation theorem. The geometric realisation of the operator pipeline on contact manifolds was developed in *Principia Orthogona, Volume Two: Contact Realisation of Generative Transitions* [2] (HAL hal-05559997v1 and hal-05565024v1, both deposited 24 March 2026).

The present paper demonstrates a multi-agent biological application of the same operator algebra. It is directly related to the following Zenodo deposits by the author:

- [5]: Zenodo 19162013 (The G6 Crystal — architectural application)
- [7]: Zenodo 19122168 (dm³ operator, explicit toy model)
- [6]: Zenodo 19117400 (dm³ operator, global dynamical analysis)

All mathematical claims in this paper follow strictly from previously proved results. The companion paper [3] specialises to the fruit-fly connectome and adds code and formal verification.

2 Operator Pipeline in Biological Systems

Definition 1 (Biological Agent Dynamics [1]). *Let each biological agent (neuron, endocrine cell, immune cell, protein domain) follow a trajectory $x_i(t)$ in a Riemannian manifold (M, g) . The collective biological state is the tuple $(x_1, \dots, x_N)$. Each agent undergoes a generative transition governed by the composite operator*

$$G = U \circ F \circ K \circ C,$$

where:

- C : local orthogonal compression (neighbourhood averaging);
- K : curvature intensification (alignment and clipping);
- F : loss of injectivity (folding of local information);
- U : stabilisation (global coherence).

The contraction-mapping theorem of [1] ensures that for sufficiently small coupling between agents, the collective system converges to a unique fixed point. Formal Lean 4/Mathlib4 verification of the arithmetic spine of this result is in the AXLE repository [8], file `MultiOrbitBioSwarm.lean`.

3 Fruit Fly Connectome as Multi-Orbit System

The FlyWire connectome [9] provides a concrete multi-agent biological system. Each neuron trajectory is modelled as an identity orbit $\mathcal{O}_i$. The full connectome is the multi-orbit system $S = \{\mathcal{O}_1, \dots, \mathcal{O}_N\}$. The modular community structure is documented in [10].

The contraction results of [1] imply that under small inter-neuron coupling, the system converges to a collective fixed point preserving local connectivity while achieving global coherence. This is a direct biological instance of multi-orbit identity stabilisation.

A specialised treatment of this setting — with Python simulation code generating four figures and Lean 4 proofs of the six-iterate convergence bound — appears in the companion paper [3].

4 HPA-Axis Stress Response as Saturated Pitchfork

The hypothalamic-pituitary-adrenal (HPA) axis exhibits threshold behaviour under acute stress [12]. This is modelled by the saturated pitchfork bifurcation proved in [1]:

$$x' = \lambda x - cx^3,$$

with clipping enforcing a bounded response. At critical coupling $|\alpha| = 0.5$, the system transitions between low-stress and high-stress attractors. The clipping function κ prevents runaway amplification and ensures recovery. Pitchfork bifurcations in coupled neural oscillators are discussed in [13].

5 Discussion

The TO/TOGT operator pipeline provides a unified dynamical-systems language for describing collective biological transitions. All mathematical claims in this paper follow directly from rigorously proved theorems in [1] and the contact-geometric realisation in [2]. The specific dm^3 numerical invariants are modelling choices matching biological scales; they are not derived from the operator algebra.

Future work will focus on explicit multi-agent simulations and experimental falsification of predicted bifurcation thresholds. The companion paper [3] already provides the simulation code and Lean 4 verification skeleton for the fruit-fly setting; the same infrastructure applies to the broader biological scope treated here.

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